Piecewise potential-vorticity inversion on the sphere

Method and implementation

`pv_inversion_spherical`

Abstract

This document describes what the package computes, the equations it solves, how those equations are discretised and solved, and what the approximations cost. It is self-contained: everything needed to follow the method is here or in the three papers listed at the end.

Contents

1 What the inversion computes

The input is two atmospheric states covering the northern hemisphere on the same pressure levels: an *event* and the *climatology* it is to be measured against. Each state is geopotential height, temperature, and the two horizontal wind components.

The output is a decomposition. The difference between the two states carries anomalous potential vorticity in the interior of the atmosphere and anomalous potential temperature on its top and bottom boundaries. Those anomalies are the *sources*. For each source separately, the method returns the balanced geopotential and streamfunction — and hence the balanced rotational wind — that this source alone induces, with every other source set to its climatological value. The pieces sum to the inversion of all the sources at once.

With the nine-level set the package uses by default, there are nine sources and therefore nine pieces:

- seven interior potential-vorticity anomalies, one on each of 850, 700, 500, 400, 300, 250 and 200 hPa;
- one potential-temperature anomaly on the bottom boundary, labelled 1000 hPa;
- one potential-temperature anomaly on the top boundary, labelled 100 hPa.

The question this answers is of the form: *of the anomalous rotational wind at 250 hPa, how much was induced by the potential vorticity anomaly at each level, and how much by the potential temperature anomaly at the ground?* The decomposition is the point. A total balanced state, taken alone, says nothing about attribution.

Three ideas underlie it.

One. Charney (1955) showed that in the motions of meteorological interest the horizontal divergence is small enough that the wind can be written as the rotational wind of a streamfunction, and that the divergence equation then collapses to a single diagnostic relation between the streamfunction and the geopotential. That relation, the nonlinear balance equation, replaces the prognostic momentum equations by a constraint (Section 5).

Two. Davis and Emanuel (1991) added the hydrostatic Ertel potential vorticity, written on Exner surfaces and with the vertical derivative of the total wind replaced by that of the rotational wind. That gives a second relation between the same two fields. The two together are a closed system: given the potential vorticity in the interior and the potential temperature on the two boundaries, the geopotential and the streamfunction are determined (Section 6).

Three. The system is nonlinear, so the response to a sum of sources is not in general the sum of the responses. Davis and Emanuel (1991) removed the ambiguity by freezing the coefficients of the operator at the mean plus half the perturbation. Davis (1992) named this the *full linear* method, compared it against the alternatives, and showed that it is the one whose pieces sum to the total by construction (Section 7).

Everything after that is the sphere. This method inverts on the whole globe in spherical harmonics. There is no lateral boundary anywhere, so there is no lateral boundary condition to choose and no piece carrying the response to one; the poles are ordinary points; and an event centred at 79 N is inverted by exactly the same calculation as one at 43 N.

Conventions

Fields are ordered bottom-up in the vertical (1000 hPa first), latitudes ascend from the equator, longitudes ascend from zero. Temperature is supplied as temperature, not potential temperature. Everything inside the solver is SI.

symbol	meaning
λ, φ	longitude, latitude
a	Earth radius, 6.371×10^6 m
Ω	Earth angular velocity, 7.292115×10^{-5} s^{-1}
$f = 2\Omega \sin \varphi$	Coriolis parameter; f^* is its floored form (Section 9)
$p,\ \Pi$	pressure; Exner function, the vertical coordinate
θ	potential temperature
Φ	geopotential
ψ	streamfunction of the rotational wind, $\mathbf{V} = \mathbf{k} \times \nabla\psi$
$\zeta = \nabla^2\psi$	relative vorticity
q	Ertel potential vorticity, SI
$\hat{q}$	the same, rescaled to the right-hand side of the inversion
$\overline{(\ )},\ (\ )'$	climatological mean, anomaly
$(\ )_n$	the nth piece
$\tilde{(\ )} = \overline{(\ )} + \frac{1}{2}(\ )'$	the state the operator is frozen at

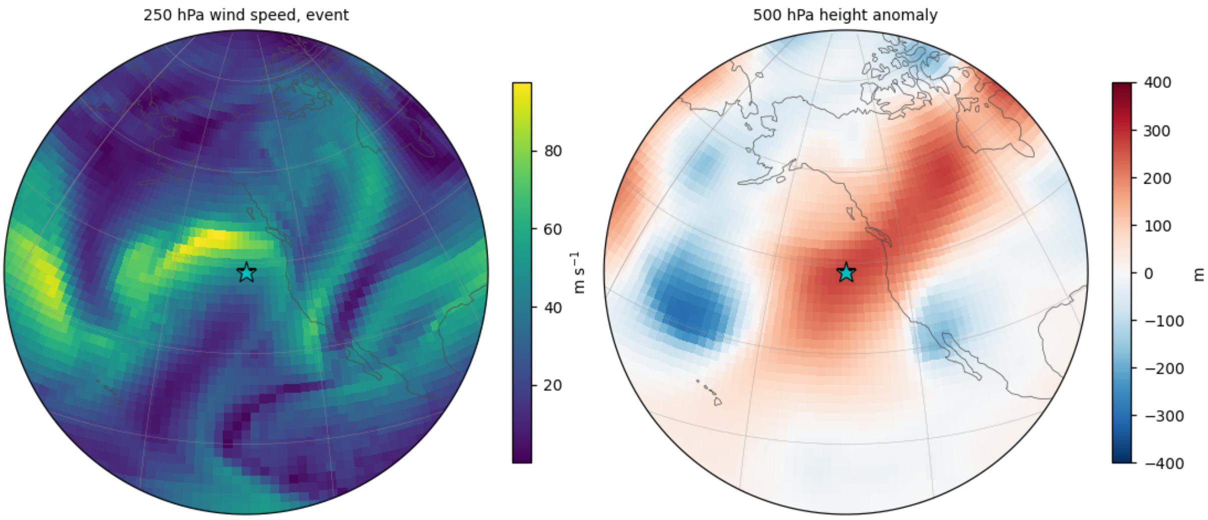

Figure 1: The event and the climatology it is measured against. The difference between the two states is what is decomposed: anomalous potential vorticity in the interior, anomalous potential temperature on the top and bottom boundaries.

2 The vertical coordinate

The vertical coordinate is the Exner function,

$$\Pi = C_p \left(\frac{p}{p_0}\right)^{\kappa}, \qquad \kappa = \frac{R_d}{C_p} = \frac{2}{7}, \tag{1}$$

with $C_p = 1004.5$ J kg^{-1} K^{-1}, $R_d = 287.0$ J kg^{-1} K^{-1} and $p_0 = 1000$ hPa. This is the coordinate of Davis and Emanuel (1991, p. 1934) and of Davis (1992, p. 1397). Note that Π *decreases* upward, and that $\Pi = C_p$ exactly at 1000 hPa.

Two things follow from (1), and they are the reason for choosing it.

Hydrostatic balance is linear in the potential temperature:

$$\frac{\partial \Phi}{\partial \Pi} = -\theta. \tag{2}$$

Neither Davis paper displays (2) as a numbered equation. It appears in both as the condition on the top and bottom boundaries (Davis and Emanuel 1991, p. 1934; Davis 1992, p. 1398), and it is used silently inside the potential-vorticity relation, where $\partial\theta/\partial\Pi$ has been replaced by $-\partial^2\Phi/\partial\Pi^2$.

One vertical operator serves both purposes. Differentiating (2) once gives the static stability,

$$S \;=\; -\frac{\partial\theta}{\partial\Pi} \;=\; \frac{\partial^2\Phi}{\partial\Pi^2}, \tag{3}$$

which is the coefficient multiplying the absolute vorticity in the potential-vorticity relation of Section 4. So the same three-point second difference in Π supplies the stability inside the equation and the boundary condition at its ends; they cannot be discretised inconsistently because they are the same stencil (Section 8).

The levels

The default set is nine pressure levels. The two outermost are *not* unknowns: they carry boundary potential temperature, and the seven between them carry potential vorticity. The boundary values are layer averages,

$$\theta_B = \tfrac{1}{2}\left(\theta_{1000} + \theta_{850}\right), \qquad \theta_T = \tfrac{1}{2}\left(\theta_{200} + \theta_{100}\right), \tag{4}$$

which places them near 925 and near 140 hPa. Davis and Emanuel (1991, p. 1936) do the same thing: their lower boundary temperature is the mean between 1000 and 850 mb, their upper boundary value sits near 125 mb, and their potential vorticity is carried on the levels in between.

p [hPa]	Π [J kg^{-1} K^{-1}]	carries	δ_k [J kg^{-1} K^{-1}]	$p/(g\kappa\Pi)$
1000	1004.5000	boundary θ	—	—
850	958.9234	interior q	−48.66	31.63
700	907.1774	interior q	−67.45	27.53
500	824.0269	interior q	−67.02	21.65
400	773.1305	interior q	−55.95	18.46
300	712.1246	interior q	−48.58	15.03
250	675.9784	interior q	−38.95	13.20
200	634.2263	interior q	−77.85	11.25
100	520.2782	boundary θ	—	—

Here $\delta_k = \frac{1}{2}(\Pi_{k+1} - \Pi_{k-1})$ is the half-span of the vertical stencil at level k, negative because Π descends; it sets the size of the boundary source in Section 8. A ten-level set is also available, inserting 150 hPa ($\Pi = 584.1810$) between 200 and 100 hPa.

The mesh is strongly stretched — the span at 200 hPa is twice that at 250 hPa, because the level above it is 100 hPa — and the consequences are stated where they bite: the second difference is exact for quadratics in Π but only formally first-order on a mesh like this one (Section 10).

Scaling of the right-hand side

Everything inside the solver is SI. The potential-vorticity equation is divided through by the factor $g\kappa\Pi/p$ that stands in front of it, so the field that actually appears on the right-hand side is

$$\hat{q} \;=\; q\,\frac{p}{g\kappa\Pi} \;=\; q\left(\frac{p}{p_0}\right)^{1-\kappa}\frac{p_0}{\kappa\, g\, C_p}, \tag{5}$$

whose per-level values are the last column of the table. The whole of the non-dimensionalisation is this one factor; no other rescaling is applied anywhere.

For plotting, a pseudoheight is convenient:

$$z = \frac{p_0}{\rho_0 \kappa g}\left[1 - \frac{\Pi}{C_p}\right] \qquad \text{(Davis and Emanuel 1991, Fig. 5)}, \tag{6}$$

$$z = H\left(1 - \frac{\Pi}{\Pi_0}\right), \quad H = \frac{C_p \theta_0}{g} \qquad \text{(Davis 1992, Fig. 2)}. \tag{7}$$

Both are linear in Π and neither enters the calculation.

3 The streamfunction

Charney's first step is to argue the divergence away. For quasi-geostrophic flow, the ratio of the horizontal divergence to one of its component terms is the Rossby number and therefore small; where the Rossby number is not small, the ratio is instead $V^2/(gH^2\,\partial \ln\theta/\partial z)$, which for $H \sim 10^4$ m and $\partial \ln\theta/\partial z \sim 10^{-5}$ m^{-1} is again small, since the gravity-inertia speed is of order 100 m s^{-1} (Charney 1955, p. 23). The significant motions of the free atmosphere therefore have small horizontal divergence, and the wind may be written through a streamfunction,

$$\mathbf{V} = \mathbf{k} \times \nabla\psi, \qquad \zeta = \nabla^2\psi. \tag{8}$$

Recovering ψ from an observed wind is, on the sphere, one spectral operation. The vorticity is formed from the wind directly; then, because the spherical harmonic of total wavenumber n is an eigenfunction of the Laplacian with eigenvalue $-n(n+1)/a^2$, the streamfunction is obtained coefficient by coefficient:

$$\psi_{nm} = \frac{\zeta_{nm}}{-n(n+1)/a^2}, \qquad n \geq 1. \tag{9}$$

There is no relaxation and no boundary value to supply, because there is no boundary. The only mode without an inverse is $n = 0$, the global constant, which carries no wind and is set to zero.

It is worth writing the vorticity out, because this is the first place the sphere's geometry enters. Define the two derivative operators

$$\partial_x = \frac{1}{a\cos\varphi}\frac{\partial}{\partial\lambda}, \qquad \partial_y = \frac{1}{a}\frac{\partial}{\partial\varphi}. \tag{10}$$

Then for a vector field with components (F_x, F_y),

$$\nabla \cdot \mathbf{F} = \partial_x F_x + \partial_y F_y - \frac{\tan\varphi}{a} F_y, \tag{11}$$

$$\mathbf{k} \cdot \nabla \times \mathbf{F} = \partial_x F_y - \partial_y F_x + \frac{\tan\varphi}{a} F_x, \tag{12}$$

$$\nabla^2 s = \partial_x\partial_x s + \partial_y\partial_y s - \frac{\tan\varphi}{a}\partial_y s. \tag{13}$$

The tan(latitude) terms are the sphere's. Dropping the one in (12) from a vorticity, for example, corrupts the potential vorticity that is built on it.

The package does not evaluate $\tan\varphi$ on the path that matters. The divergence and the vorticity are formed by transforming products of grid fields against the derivatives of the Legendre functions — an integration by parts in latitude — so the metric appears only as $1/(1 - \sin^2\varphi)$ inside an integral over a grid that has no row at the pole. The gradient is synthesised from a table of Legendre combinations built by recursion rather than by differentiation, so it too stays finite there. This matters at high latitude, where $\tan\varphi$ does not.

The solver grid is a Gauss–Legendre grid with an *even* number of latitude rows, so no row lands on the equator, and it has no pole row. Both facts are used: the equator is where the mirror of Section 9 is joined, and the pole is where $\cos\varphi$ would otherwise be divided by.

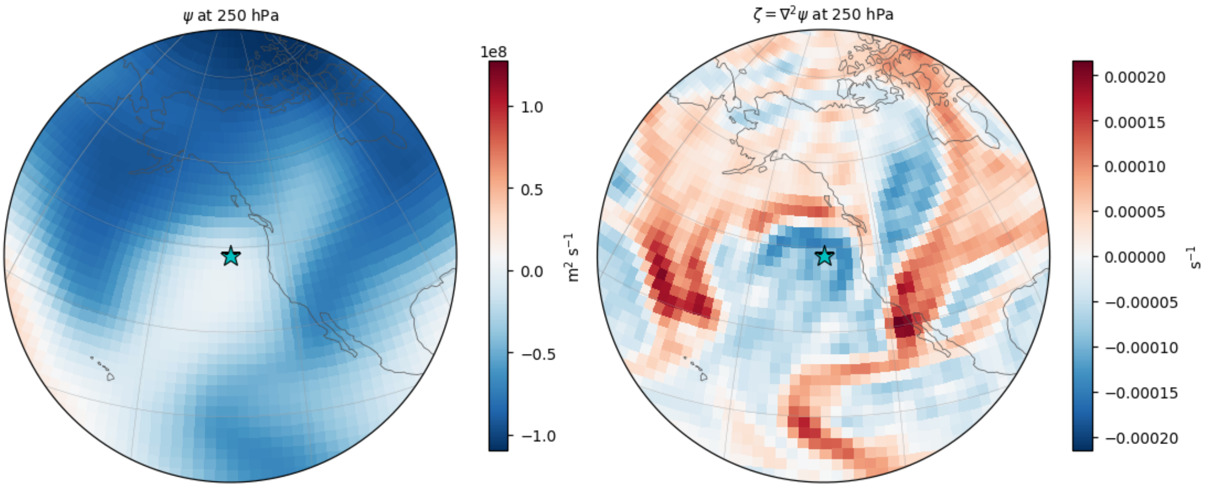

Figure 2: The streamfunction and the vorticity it carries. Left: the streamfunction recovered from the observed wind. Right: its Laplacian, the relative vorticity, formed by quadrature rather than by differencing so that the tan(latitude) term is carried without being evaluated.

4 Potential vorticity

Ertel's potential vorticity is

$$q = \frac{1}{\rho}\boldsymbol{\eta}\cdot\nabla\theta, \tag{14}$$

with ρ the density, $\boldsymbol{\eta}$ the absolute vorticity vector, θ the potential temperature and the gradient three-dimensional. It is conserved following three-dimensional, adiabatic, inviscid motion. This is (1.1) of Davis and Emanuel (1991) and (1.2) of Davis (1992).

Making the hydrostatic approximation and writing (14) on Exner surfaces in spherical coordinates gives (2.2) of Davis and Emanuel (1991),

$$q = -\frac{g\kappa\Pi}{p}\left(\eta\frac{\partial\theta}{\partial\Pi} - \frac{1}{a\cos\varphi}\frac{\partial v}{\partial\Pi}\frac{\partial\theta}{\partial\lambda} + \frac{1}{a}\frac{\partial u}{\partial\Pi}\frac{\partial\theta}{\partial\varphi}\right), \tag{15}$$

where η is now the *vertical component* of the absolute vorticity — the same symbol that stands for the vector in (14), in the same paper. Below, $f + \nabla^2\psi$ is written out instead.

Two substitutions turn (15) into a relation between ψ and Φ alone. The vertical derivative of the total wind is replaced by the vertical derivative of the rotational wind, and θ is eliminated with the hydrostatic relation (2), so that $\partial\theta/\partial\Pi = -\partial^2\Phi/\partial\Pi^2$. The result is (2.3) of Davis and Emanuel (1991),

$$q = \frac{g\kappa\Pi}{p}\left[(f+\nabla^2\psi)\frac{\partial^2\Phi}{\partial\Pi^2} - \frac{1}{a^2\cos^2\varphi}\frac{\partial^2\psi}{\partial\lambda\,\partial\Pi}\frac{\partial^2\Phi}{\partial\lambda\,\partial\Pi} - \frac{1}{a^2}\frac{\partial^2\psi}{\partial\varphi\,\partial\Pi}\frac{\partial^2\Phi}{\partial\varphi\,\partial\Pi}\right]. \tag{16}$$

Its Cartesian counterpart is (1.4) of Davis (1992), whose last two terms are written together as $L_2(A,B) = -A_{x\Pi}B_{x\Pi} - A_{y\Pi}B_{y\Pi}$.

The two metric factors in (16) are exactly what is needed to make those last two terms one horizontal dot product. With $\partial_x = (a\cos\varphi)^{-1}\partial_\lambda$ and $\partial_y = a^{-1}\partial_\varphi$ as in Section 3, and with the scaling (5), the relation the package builds is

$$\hat{q} = (f^* + \nabla^2\psi)\frac{\partial^2\Phi}{\partial\Pi^2} - \nabla\left(\frac{\partial\psi}{\partial\Pi}\right)\cdot\nabla\left(\frac{\partial\Phi}{\partial\Pi}\right). \tag{17}$$

The first term is absolute vorticity times static stability. The second measures the correlation between the vertical shear of the rotational wind and the horizontal gradient of potential temperature, since $\nabla(\partial\Phi/\partial\Pi) = -\nabla\theta$. Both are formed as pointwise products on the grid and transformed back, so no metric factor is ever evaluated.

The unit is the potential-vorticity unit, 1 PVU = 10^{-6} m^2 K kg^{-1} s^{-1}. A typical tropospheric value is 0.5 PVU and stratospheric values are roughly an order of magnitude larger; the tropopause is conventionally drawn at the 1.5 PVU contour (Davis and Emanuel 1991, p. 1939 and footnote 8).

Positivity is not cosmetic. Equation (17) is elliptic only where the absolute vorticity and the static stability are both positive, which is the same as $q > 0$ where the cross term is small. Davis and Emanuel (1991, Appendix A) report that their solution did not converge unless that condition was imposed on the observed field, and set negative values to 10^{-2} PVU. This package floors the potential vorticity too, and gives the added amount back by area weight; the values and the accounting are in Section 15.

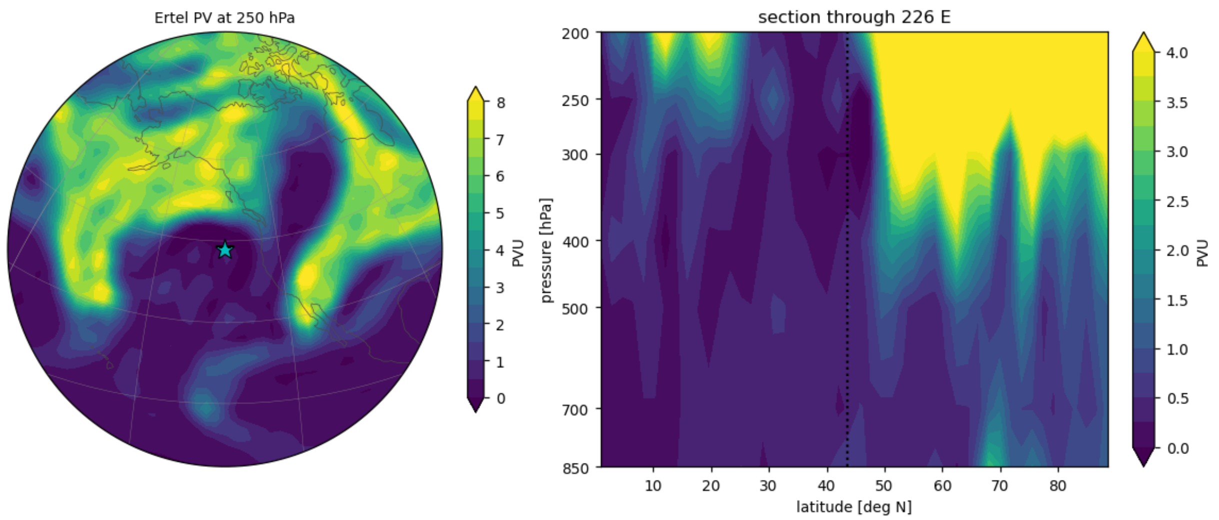

Figure 3: Ertel potential vorticity for the event. Left: a map on one level. Right: a vertical section through the anomaly, with the tropopause visible as the crowding of the contours.

5 The balance equation

5.1 As Charney wrote it

Charney (1955, p. 24) obtains the balance equation by omitting the small vertical-advection terms, substituting the streamfunction for the wind, and taking the horizontal divergence of the equations of motion with pressure as the vertical coordinate. As printed, and with ϕ the geopotential, it reads

$$\nabla^2\psi + \nabla\psi\cdot\nabla f - \frac{2}{f}\left[\left(\frac{\partial^2\psi}{\partial x\,\partial y}\right)^2 - \frac{\partial^2\psi}{\partial x^2}\frac{\partial^2\psi}{\partial y^2}\right] = \frac{\nabla^2\phi}{f}. \tag{18}$$

The paper carries no equation numbers, so (18) is cited by page alone. One transcription point: every term of (18) except $\nabla\psi\cdot\nabla f$ has been divided through by f, and as printed that term is dimensionally inconsistent with the rest. Read as $\nabla\psi\cdot\nabla f/f$ and multiplied through by f, the equation becomes the form used everywhere below.

Charney gives the condition under which (18) can be solved for ψ as a boundary-value problem:

$$\frac{\nabla^2\phi}{f} + \frac{f}{2} > 0, \tag{19}$$

that is, the geostrophic relative vorticity must exceed $-f/2$. He also notes that if the nonlinear terms are small the relation integrates to the geostrophic approximation, so the balance equation is a generalisation of it: both determine the velocity from the geopotential.

Multiplied through by f and rearranged, (18) is (1.3) of Davis (1992),

$$\nabla^2\Phi = \nabla \cdot f\nabla\psi + 2\left[\frac{\partial^2\psi}{\partial x^2}\frac{\partial^2\psi}{\partial y^2} - \left(\frac{\partial^2\psi}{\partial x\,\partial y}\right)^2\right], \tag{20}$$

and in spherical coordinates it is (2.1) of Davis and Emanuel (1991),

$$\nabla^2\Phi = \nabla \cdot (f\nabla\psi) + \frac{2}{a^4\cos^2\varphi}\,\frac{\partial(\partial\psi/\partial\lambda,\ \partial\psi/\partial\varphi)}{\partial(\lambda,\varphi)}. \tag{21}$$

The a^4 is not a misprint: it is what turns the Jacobian in (λ,φ) into the determinant of the second derivatives in the local x and y.

The assumption behind it is stated in both papers. It is obtained by taking the horizontal divergence of the momentum equations, splitting the wind into rotational and irrotational parts, and retaining terms of order R_ψ while neglecting those of order R_χ and higher, where R_ψ and R_χ are Rossby numbers built on the two parts of the wind (Davis and Emanuel 1991, p. 1934). It is accurate in flows with large curvature because it is close to gradient-wind balance, and it remains accurate as long as the Froude number is small even when the Rossby number is large (Davis 1992, p. 1398). Charney's own justification for dropping the divergence terms is empirical: they are one to two orders of magnitude smaller than the rest.

5.2 Where the sphere enters

Two distinct pieces of geometry appear, and they are worth separating.

The tan(latitude) terms. They are already inside (21): neither the spherical divergence nor the spherical Laplacian is the flat one. Writing them out with the operators (11) and (13) of Section 3,

$$\nabla \cdot (f\nabla\psi) = f\left[\partial_x\partial_x\psi + \partial_y\partial_y\psi - \frac{\tan\varphi}{a}\,\partial_y\psi\right] + \beta\,\partial_y\psi, \qquad \beta = \frac{1}{a}\frac{\partial f}{\partial\varphi} = \frac{2\Omega\cos\varphi}{a}. \tag{22}$$

The tan(latitude) term is not a small correction to the β term. Their ratio is

$$\frac{f\tan\varphi/a}{\beta} = \tan^2\varphi, \tag{23}$$

so at 45 N the two are exactly equal, at 60 N the tan(latitude) term is three times as large, and poleward of that it dominates. Dropping it does not perturb the balance equation; it changes which equation is being solved.

The curvature term. The package does not evaluate the determinant of second derivatives at all. It solves the balance equation in the form

$$\nabla^2\Phi = N(\psi), \qquad N(\psi) = \nabla \cdot [(f^* + \zeta)\,\nabla\psi] - \tfrac{1}{2}\nabla^2\,|\nabla\psi|^2, \qquad \zeta = \nabla^2\psi, \tag{24}$$

in which every term is a divergence or a Laplacian of a product formed on the grid. Nothing here is a second derivative in latitude and nothing divides by $\cos\varphi$, so the expression is regular everywhere, poles included.

On the sphere the two forms are not the same. The identity is

$$\nabla \cdot (\zeta \nabla \psi) - \tfrac{1}{2} \nabla^2 |\nabla \psi|^2 = 2 \det(\text{Hess}\, \psi) - \frac{|\nabla \psi|^2}{a^2}, \tag{25}$$

so (24) falls short of $\nabla \cdot (f^* \nabla \psi) + 2 \det(\text{Hess}\, \psi)$ — the spherical reading of the Jacobian group of (21) — by exactly $|\nabla \psi|^2/a^2 = |\mathbf{V}|^2/a^2$. That is the sphere's curvature. It is one-signed and systematic rather than random, and for a 30 m s^{-1} flow it is about 2% of the balance terms. The package computes it and reports it as a diagnostic rather than absorbing it silently.

The determinant form is retained as an independent check, through the two deformation components. These carry the tan(latitude) terms explicitly:

$$D_1 = \partial_x u - \partial_y v - \frac{\tan \varphi}{a} v, \qquad D_2 = \partial_x v + \partial_y u + \frac{\tan \varphi}{a} u, \tag{26}$$

pairing with $\nabla \cdot \mathbf{V} = \partial_x u + \partial_y v - (\tan \varphi / a) v$ and $\zeta = \partial_x v - \partial_y u + (\tan \varphi / a) u$, and giving

$$2 \det(\text{Hess}\, \psi) = \tfrac{1}{2} \left(\zeta^2 - D_1^2 - D_2^2 \right). \tag{27}$$

This is a check and not the production path, precisely because (26) evaluates $\tan \varphi$ and therefore degrades towards the poles, while (24) does not.

The Coriolis parameter in (24) is f^*, the floored form of Section 9, not the true f. The reason is visible in (22): the balance term differentiates f, and $|f|$ on a mirrored hemisphere has a corner at the equator.

6 The coupled system

The balance equation relates Φ to ψ. The potential-vorticity relation relates them again, differently. Together they are a closed system for the two fields, given the potential vorticity in the interior and the potential temperature on the two boundaries:

$$\nabla^2 \Phi = \nabla \cdot \left[\left(f^* + \nabla^2 \psi \right) \nabla \psi \right] - \tfrac{1}{2} \nabla^2 |\nabla \psi|^2, \tag{28}$$

$$\hat{q} = \left(f^* + \nabla^2 \psi \right) \frac{\partial^2 \Phi}{\partial \Pi^2} - \nabla \left(\frac{\partial \psi}{\partial \Pi} \right) \cdot \nabla \left(\frac{\partial \Phi}{\partial \Pi} \right), \tag{29}$$

with $\partial \Phi / \partial \Pi = -\theta$ at the top and the bottom (Section 8). Davis and Emanuel (1991, p. 1934) state it in exactly these terms: their (2.1) and (2.3), which are (28) and (29) here, "form a complete system for the unknowns Φ and Ψ, given q".

Three properties of the system decide everything that follows.

It is elliptic only conditionally. Both equations are elliptic where the absolute vorticity $f^* + \nabla^2 \psi$ and the static stability $\partial^2 \Phi / \partial \Pi^2$ are positive (Davis and Emanuel 1991, Appendix A). Where either changes sign the character of the problem changes with it, and no amount of iteration will recover a solution. The floors of Section 15 exist for this reason and for no other.

It is nonlinear, but only mildly. Equation (28) is quadratic in ψ. Equation (29) is bilinear in (ψ, Φ): every term is one factor of ψ times one factor of Φ, and the coefficients depend on neither. No term is of higher degree than two, and there is no transcendental dependence anywhere. Section 7 is built on that fact.

The response to a sum of sources is not the sum of the responses. That is what nonlinearity means here, and it is the whole difficulty of doing the inversion piecewise. Two potential-vorticity anomalies inverted together do not give the sum of what each gives alone, because each changes the coefficients the other is inverted against.

The total state, and the mean it is measured from

The system (28)–(29) is solved once, for the event, by Newton's method (Section 10). That gives the balanced total state $(\Phi_{\text{bal}}, \psi_{\text{bal}})$. The perturbation that the piecewise operator is frozen at is then

$$\psi' = \psi_{\text{bal}} - \bar{\psi}, \qquad \Phi' = \Phi_{\text{bal}} - \bar{\Phi}, \tag{30}$$

measured from the *observed* climatological mean rather than from a separately balanced one.

That is a choice, and it has a cost. Whatever departure from balance the observed mean carries is absorbed into ψ' and Φ'. Davis (1992, p. 1411) describes the same shortcut in its stronger form — assuming the observed mass and rotational wind fields are in balance and proceeding directly to the piecewise inversion — and notes that the discrepancy this introduces is of the order of the departure of those fields from balance, which is generally small. Here only the mean is taken as observed; the event is balanced properly. The consequence appears in the physical residual of Section 13, never in the closure of the pieces.

7 The piecewise linearisation

7.1 The choice, and why it is the one that closes

Davis and Emanuel (1991, p. 1938) put the problem and their answer in one sentence: linear equations are needed for a piecewise inversion, and rather than dispensing with the nonlinear terms altogether, the linearisation is performed by hiding the nonlinear terms in the non-constant coefficients of a linear differential operator.

The algebra is clearest on a single product. Let $q = AB$, split each field into a mean and an anomaly, and split each anomaly into N pieces. Then

$$q' = \bar{A}B' + \bar{B}A' + A'B', \tag{31}$$

which is (2.5) of Davis and Emanuel (1991), and summing the pieces,

$$\sum_{n=1}^{N} q_n = \bar{A}\sum_n B_n + \bar{B}\sum_n A_n + A_1B_1 + A_1B_2 + A_2B_1 + \cdots + A_NB_N. \tag{32}$$

The difficulty is entirely in the last group. The linear terms of (32) assign themselves to pieces; the quadratic ones belong to pairs of pieces and to no single one. Two extreme assignments give the whole of the quadratic term to A_n or the whole of it to B_n — (2.7) and (2.8) of the same paper — and neither is preferable to the other. Davis and Emanuel take their average,

$$q_n = \left(\bar{A} + \tfrac{1}{2}\sum_{n=1}^{N} A_n\right) B_n + \left(\bar{B} + \tfrac{1}{2}\sum_{n=1}^{N} B_n\right) A_n, \tag{33}$$

their (2.9), on the grounds that it is the only choice invariant under exchanging A and B. Writing $\tilde{(\,)} = \bar{(\,)} + \frac{1}{2}(\,)'$, equation (33) is simply $q_n = \tilde{A}B_n + \tilde{B}A_n$.

Davis (1992) named this the *full linear* method and set it beside three others: truncated linear, which drops the quadratic term instead of splitting it; subtraction of the piece from the total followed by a fresh nonlinear solve; and addition of the piece to the mean. The differences between them are generally less than 25% even at large Rossby number, with truncated linear the outlier; accounting for the nonlinear terms in the balance equation matters. Davis stresses that there is no unambiguously correct method and that the right one depends on the problem.

For this package the deciding property is the one stated in the conclusions of Davis (1992, pp. 1410–1411): the nonlinear methods have the disadvantage that the sum of the solutions for each anomaly does not equal the total perturbation flow, whereas the full linear method is defined so that it does. That is what allows the inversion to be read as a decomposition at all.

7.2 The linear system

Applying (33) to (28) and (29) gives, for the piece driven by the source $(\hat{q}_n, \theta_{B,n}, \theta_{T,n})$, a *linear* system for that piece's (Φ_n, ψ_n):

$$\nabla^2\Phi_n = \nabla\cdot\left[\left(f^* + \tilde{\zeta}\right)\nabla\psi_n\right] + \nabla\cdot\left[\left(\nabla^2\psi_n\right)\nabla\tilde{\psi}\right] - \nabla^2\left(\nabla\tilde{\psi}\cdot\nabla\psi_n\right), \tag{34}$$

$$\hat{q}_n = A\,\frac{\partial^2\Phi_n}{\partial\Pi^2} + S\,\nabla^2\psi_n - \nabla\left(\frac{\partial\tilde{\psi}}{\partial\Pi}\right)\cdot\nabla\left(\frac{\partial\Phi_n}{\partial\Pi}\right) - \nabla\left(\frac{\partial\tilde{\Phi}}{\partial\Pi}\right)\cdot\nabla\left(\frac{\partial\psi_n}{\partial\Pi}\right), \tag{35}$$

with

$$\tilde{\psi} = \bar{\psi} + \tfrac{1}{2}\psi', \qquad \tilde{\Phi} = \bar{\Phi} + \tfrac{1}{2}\Phi', \qquad \tilde{\zeta} = \nabla^2\tilde{\psi}, \qquad A = f^* + \tilde{\zeta}, \qquad S = \frac{\partial^2\tilde{\Phi}}{\partial\Pi^2}. \tag{36}$$

Equations (34) and (35) are (2.10) and (2.11) of Davis and Emanuel (1991), and (3.1) and (3.2) of Davis (1992), written in the divergence form of Section 5 and on the sphere.

Everything frozen is frozen once, before any piece is solved:

coefficient	definition	where it acts
f^*	floored Coriolis parameter — fixed, not a linearisation	both rows
$\tilde{\psi}$, $\tilde{\zeta}$	$\bar{\psi} + \frac{1}{2}\psi'$ and its Laplacian	balance row
A	$f^* + \tilde{\zeta}$, absolute vorticity, floored	potential-vorticity row
S	$\partial^2\tilde{\Phi}/\partial\Pi^2$, static stability, floored	potential-vorticity row
$\nabla(\partial\tilde{\psi}/\partial\Pi)$	reference shear of the streamfunction	cross terms
$\nabla(\partial\tilde{\Phi}/\partial\Pi)$	reference shear of the geopotential	cross terms

The two reference vertical derivatives are centred over the full set of levels, including the two outermost ones. No ghost is needed for them, because $\bar{\Phi}$ and $\bar{\psi}$ are known everywhere; only the unknown needs the boundary treatment of Section 8.

7.3 Why the linearisation is exact

The half in $\tilde{(\ )}$ is usually presented as a symmetry argument. On a system of degree two it is more than that: it makes the linearisation an identity.

Write the balance nonlinearity as $N(\psi) = Q(\psi,\psi)$ plus linear terms, with Q a symmetric bilinear form, and write the potential-vorticity cross term as a bilinear $B(\psi,\Phi)$. Then, for any mean state and any perturbation,

$$N(\bar{x} + x') - N(\bar{x}) = DN\big(\bar{x} + \tfrac{1}{2}x'\big)\big[x'\big], \tag{37}$$

$$B(\bar{x} + x',\, \bar{y} + y') - B(\bar{x},\bar{y}) = B\big(x',\, \bar{y} + \tfrac{1}{2}y'\big) + B\big(\bar{x} + \tfrac{1}{2}x',\, y'\big)\,. \tag{38}$$

Both lines are identities, not first-order truncations. Expanding (37): the left side is $2Q(\bar{x},x') + Q(x',x')$, and the right side is $2Q(\bar{x} + \frac{1}{2}x',\, x')$, which is the same. Expanding (38): the left side is $B(\bar{x},y') + B(x',\bar{y}) + B(x',y')$, and the right side is $B(x',\bar{y}) + \frac{1}{2}B(x',y') + B(\bar{x},y') + \frac{1}{2}B(x',y')$, again the same.

Equations (34) and (35) realise both. In (35) the term $A\,\partial^2\Phi_n/\partial\Pi^2$ supplies $B\big(\bar{\psi} + \frac{1}{2}\psi',\, \Phi_n\big)$ and the term $S\nabla^2\psi_n$ supplies $B\big(\psi_n,\, \bar{\Phi} + \frac{1}{2}\Phi'\big)$, which is (38) term for term; in (34) the three terms are $DN(\tilde{\psi})[\psi_n]$.

So the operator is not a tangent approximation to the system. It is the exact finite-increment operator: applied to the full perturbation, it returns exactly the difference between the nonlinear system evaluated at the total state and at the mean state. That is what makes the piece amplitudes quantitative rather than indicative. The one qualification — that this is exactness with respect to the operator that is actually built, and that the floors and the equatorial taper modify that operator — is set out in Section 15.

7.4 What a piece is

One piece per source. With the nine-level set there are nine: the bottom boundary temperature, the seven interior potential-vorticity levels, and the top boundary temperature. Every level index belongs to exactly one piece, and the list is checked rather than assumed.

The sources are taken exactly as Davis and Emanuel (1991, p. 1940) take them: either $\hat{q}_n$ is the anomaly on one interior level with $\theta_n = 0$ on both boundaries, or θ_n is the anomaly on one boundary with $\hat{q}_n = 0$ throughout the interior. Nothing else is ever nonzero.

The potential-vorticity anomaly of a piece is formed by flooring the event field and the mean field *separately* and then differencing them; the boundary temperature anomalies are plain differences. The flooring is not linear, so this is the anomaly of two floored fields and not the floored anomaly. The ordering is deliberate and the reasoning is in Section 15.

There is no piece for a lateral boundary, because there is no lateral boundary (Section 8).

7.5 Why the pieces add exactly

Four facts, each of which can be checked independently.

One operator. The frozen coefficients, the assembled operator and its preconditioner are built once, outside the loop over pieces. Every piece is one solve against the same M. The floors and the equatorial taper are baked into it identically for all pieces — which is why the taper is applied to coefficients and never to a source or to a solution.

The right-hand side is linear in the source. The balance row's right-hand side is identically zero apart from the gauge slot of Section 10. The potential-vorticity row's is

$$b_2 = \hat{q}_n - A\,\Theta(\theta_{B,n}, \theta_{T,n}) + \nabla\left(\frac{\partial\tilde{\psi}}{\partial\Pi}\right)\cdot\nabla\Phi_n^{\mathrm{gh}} + \nabla\left(\frac{\partial\tilde{\Phi}}{\partial\Pi}\right)\cdot\nabla\psi_n^{\mathrm{gh}}, \tag{39}$$

where Θ is the folded boundary source of Section 8 and the two ghost contributions are the boundary parts of the first differences in Π inside the cross terms. Every one of those maps is linear in $(\hat{q}_n, \theta_{B,n}, \theta_{T,n})$: Θ is linear, the ghost of a first difference is linear, and the streamfunction ghost (43) is linear, being an area-mean removal followed by division by a fixed f^*.

The sources partition exactly. Each interior level's anomaly goes to one piece and each boundary temperature to one piece, and their sum is the whole anomaly. This is why a group defined by something other than a level list has to be an additive split of the *source*: a multiplicative mask applied to a solution partitions nothing (Section 12).

The inverse is linear. $\sum_n M^{-1}b_n = M^{-1}\sum_n b_n$, which is the solve with all sources present at once. The hydrostatic reconstruction of the two boundary levels is affine with a linear temperature term, so the extended nine-level arrays sum as well.

Put together with the exactness of (37)–(38), the object the pieces sum to is the exact perturbation the system implies, not a tangent approximation to it. Two limits on that statement are worth carrying forward. The closure is exact in exact arithmetic; in practice each piece is solved to a finite tolerance and the closure is measured rather than assumed. And the pieces sum to the all-sources *linear* solve, not to $\psi_{\text{bal}} - \bar{\psi}$. Both are the subject of Section 13.

8 Boundary conditions

8.1 Top and bottom

The condition at the top and the bottom is hydrostatic balance,

$$\frac{\partial \Phi}{\partial \Pi} = -\theta \qquad \text{at } \Pi = \Pi_0 \text{ and } \Pi = \Pi_T, \tag{40}$$

and for a piece, $\partial\Phi_n/\partial\Pi = -\theta_n$ (Davis and Emanuel 1991, p. 1939; Davis 1992, p. 1400). It is implemented as a pair of ghost values just outside the interior:

$$\Phi_0 = \Phi_1 + \theta_B(\Pi_1 - \Pi_0), \qquad \Phi_{N-1} = \Phi_{N-2} - \theta_T(\Pi_{N-1} - \Pi_{N-2}). \tag{41}$$

Substituting (41) into the end rows of the vertical second difference splits them into two parts: a homogeneous part, in which one coupling weight is folded onto the diagonal, and a source, in which the boundary temperature moves to the right-hand side. With the three-point weights of Section 10 the source is

$$\text{contribution to } \frac{\partial^2 \Phi}{\partial \Pi^2}\bigg|_{k=1} = \frac{\theta_B}{\delta_1}, \qquad \text{at } k = N-2: \ -\frac{\theta_T}{\delta_{N-2}}, \qquad \delta_k = \tfrac{1}{2}(\Pi_{k+1} - \Pi_{k-1}), \tag{42}$$

which is consistent with the folding because $b_k^{\text{lo}}(\Pi_k - \Pi_{k-1}) = b_k^{\text{hi}}(\Pi_{k+1} - \Pi_k) = 1/\delta_k$ identically. Since Π decreases upward, $\delta_k < 0$, so a warm lower boundary lowers the static-stability term at the level above it — which on the right-hand side acts like a positive potential-vorticity source, as it should.

Keeping the folding and the source apart is what allows a piece to be driven by one boundary temperature alone: the operator is shared, only the source changes.

The boundary temperature enters *twice*, and the second place is easy to miss. Besides the static-stability term, the cross terms of (35) contain first differences in Π, and a centred first difference at the first interior level reaches the ghost too. Both contributions are assembled from one call to the vertical operator so that they cannot drift apart.

8.2 The streamfunction at the boundaries

Hydrostatic balance fixes the geopotential. It says nothing directly about the streamfunction, so a ghost for ψ is a statement about the flow just outside the domain, and the defensible statement is that it is balanced there. Davis and Emanuel (1991) apply $\partial\psi/\partial\Pi = -\theta$ at both boundaries and justify it by noting that the term involving $\partial\psi/\partial\Pi$ in (29) is already small, so a correction accounting for nonlinear balance would enter at second order; their solutions are, they report, fairly insensitive to the choice. Davis (1992) writes the pair together as $\partial\Phi/\partial\Pi = f_0\,\partial\psi/\partial\Pi = -\theta$.

With f varying, this package uses

$$\frac{\partial \psi}{\partial \Pi} = -\frac{\theta - \langle\theta\rangle}{f^*}, \tag{43}$$

where $\langle\cdot\rangle$ is the area mean. Two details in (43) are load-bearing.

The division by f^*, because to leading order $\Phi = f\psi$ and without it the expression is not dimensionally a streamfunction at all. It is performed *before* the horizontal gradient is taken, since f varies with latitude and $\nabla(g/f) \neq \nabla(g)/f$.

The removal of the area mean, because the horizontally uniform part of the temperature carries no flow and must not reach the streamfunction. Divided by f without it, the ghost is of order 10^8 m^2 s^{-1} — larger than the streamfunction itself.

The same convention is used in three places — the right-hand side of the linear operator, the nonlinear residual, and the reconstruction of the boundary levels — because any two of them disagreeing leaves a converged state failing its own equations.

8.3 Why there is no condition at the side

On a limited-area domain the elliptic problem needs data on its edge, and the choice is awkward. Davis and Emanuel (1991) put it directly: appropriate lateral conditions are difficult to determine because the contribution of an individual anomaly to the flow at any point is not known in advance. Their approach is to choose a domain much larger than the region of interest and apply homogeneous lateral conditions to each ψ_n and Φ_n; they note the alternative of inverting over a global domain, or at least a hemisphere. Davis (1992) places the lateral boundaries more than a Rossby radius from the perturbation of interest and applies homogeneous Dirichlet conditions there.

This method takes the alternative. The domain is the whole sphere. There is no edge, so there is no lateral condition, no arbitrary distance to choose, and no piece that carries the response to an imposed wall. The decomposition is exactly the sources.

What a global domain must supply instead is a gauge. The operator has an exact null space — a constant added to ψ on any level carries no wind, and a single global constant added to Φ passes through the vertical stencil — and it must be removed explicitly rather than by a wall. Section 10 describes how.

The lower boundary in particular deserves its status as a piece in its own right. Davis (1992, p. 1407) shows that for an anomaly placed at the second lowest interior level the rigid lower boundary raises the wind maximum by almost 50%, because the isentropes cannot bow upward underneath the anomaly; lowering the bottom surface by 4 km restores the free-atmosphere answer. The boundary is not a detail of the numerics. It is part of the physics being decomposed.

9 From a hemisphere to the sphere

The data cover the northern hemisphere; the operator is global. The extension has to do two things: keep the inversion elliptic, and keep the invented hemisphere from leaking into the answer.

One construction does both. If every coefficient and every source is an *even* function of latitude, and the Coriolis parameter is replaced by an even function of latitude, then the operator commutes with the reflection $\varphi \to -\varphi$. The solution is then exactly even, and its northern half solves the northern problem under a condition of no meridional gradient at the equator. The southern half is determined by the northern one and carries no information of its own.

9.1 Two mirrors, with different parities

Getting there takes two reflections, and the order matters.

1. **Mirror the wind as a vector** — eastward component even, northward component odd — and take the vorticity of the result. This is the only parity whose mirrored vorticity agrees with the hemisphere's own.

2. **Mirror that vorticity as an even scalar**, then take $\psi = \nabla^{-2}\zeta$ and use the rotational wind of that ψ downstream.
3. **Mirror height and temperature evenly** from the outset.

Stopping after step 1 leaves ζ odd, so $|f| + \zeta$ changes sign across the equator, ellipticity is lost in the south, and the scaffold contaminates the north at order ζ/f. The wind implied by step 2 is eastward-odd and northward-even: the mirror image of a circulation with the *same* sense of rotation, which is what $|f|$ describes.

The latitude axis is handled either way round: the northern axis may start on the equator, in which case the global axis has $2n - 1$ rows, or half a step from it, in which case it has $2n$. The solver grid must be symmetric about the equator with an even number of rows.

9.2 The Coriolis floor

$|f|$ has a corner at the equator, and that corner would appear inside ∇f in the balance term. It is smoothed:

$$f^* = 2\Omega\sqrt{\sin^2\varphi + \sin^2\varphi_0}, \qquad \varphi_0 = 12^\circ \text{ by default.} \tag{44}$$

At the equator $f^* = 3.0322 \times 10^{-5}\ \mathrm{s}^{-1}$, which is exactly $|f|$ at 12 N.

9.3 The equatorial taper

An even mirror is only continuous across the equator where the mean zonal wind vanishes there, which in general it does not; what is left is a kink in $\tilde{\zeta}$. The *coefficients* are therefore tapered to their smooth equatorial limits across a band: $\tilde{\zeta} \to 0$, $S \to$ its per-level area mean, and the four cross-term gradient stacks $\to 0$. The weight is a smooth step,

$$w = x^3\left(10 - 15x + 6x^2\right), \qquad x = \operatorname{clip}\left(\frac{|\varphi| - \varphi_S}{\varphi_N - \varphi_S},\, 0,\, 1\right), \tag{45}$$

with $\varphi_S = 5^\circ$ and $\varphi_N = 20^\circ$ by default, so w and its first two derivatives are continuous at both ends.

The taper is applied to coefficients only — never to f^*, never to a source, never to a solution. That is what keeps the operator identical across pieces and leaves the exact closure of Section 7 untouched.

9.4 What both cost

Neither is a small correction in the subtropics, and both are set for the equator rather than for the region a mid-latitude event can reach:

latitude [deg N]	0.5	1	2	5	10	10.5	15	20	30
$f^*/\lvert f\rvert$	23.85	11.95	6.04	2.59	1.56	1.52	1.28	1.17	1.08
taper weight w	0	0	0	0	0.21	0.26	0.79	1.00	1.00

At 10.5 N the Coriolis parameter is half as large again as the true one, and 74% of the reference vorticity has been removed from the operator's coefficients. The ratio is still 1.04 at 45 N.

The consequence is a rule, not a caveat: **nothing about the subtropical part of a solution should be quoted as physics without re-running with a smaller Coriolis floor and a**

narrower taper, and reporting how much of it survives. A patch centred below about 42 N reaches into the band; a patch centred below the data's own southernmost row reaches into the mirrored scaffold, and those rows are returned blank rather than filled with reflected weather.

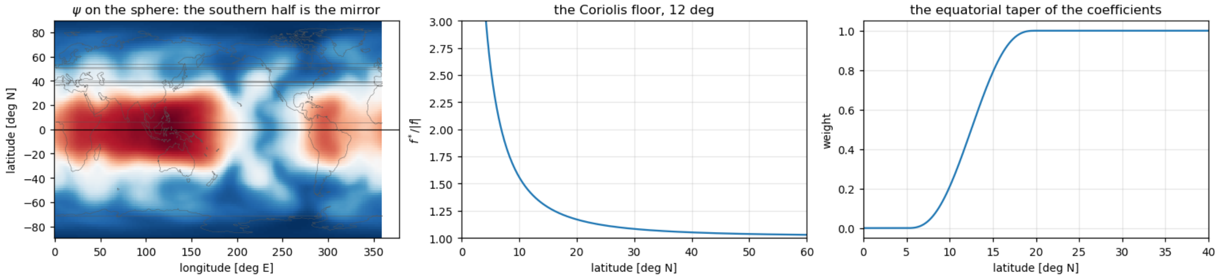

Figure 4: The hemisphere extended to the sphere. Left: the streamfunction on the global grid, with the southern half the mirror of the northern. Centre: the Coriolis floor, as the ratio $f^*/|f|$. Right: the weight applied to the operator's coefficients across the equatorial band.

10 Discretisation and solution

10.1 Two grids

Two grid roles are kept apart, and that separation is what makes the poles ordinary points.

role	grid	what happens on it
data	the file's own grid, poles included	scalar analysis and synthesis only
solver	Gauss–Legendre, no pole row	every product, coefficient and derivative

Moving between the two is exact for band-limited fields. Winds are carried across as $u\cos\varphi$ and $v\cos\varphi$, which are band limited where the bare components are not.

The solver grid is sized from the truncation L: the number of latitudes is $\lfloor(3L+1)/2\rfloor+1$, rounded up to an even number so that no row lands on the equator, and the number of longitudes is the next power of two at least twice that. The three-halves rule is deliberate — both equations are quadratic, so products of two fields truncated at L are represented without aliasing on a grid carrying $3L/2$. Read the other way, the largest truncation a grid of n_{lat} rows carries without aliasing is $2n_{\mathrm{lat}}/3-1$. Collocating instead at the largest truncation the grid can represent aliases every product and costs several times the work per solve.

10.2 Which operations are spectral and which are not

- **Coefficient multiplies:** ∇^2 (multiply by $-n(n+1)/a^2$), ∇^{-2} (divide, with $n=0$ set to zero), $\partial/\partial\lambda$ (multiply by $\mathrm{i}m$).
- **Synthesised to the grid:** the gradient. Its meridional half comes from a table of Legendre combinations built by recursion, so it is regular at the poles.
- **Formed by quadrature:** the divergence and the vorticity of a grid vector field, using the Gaussian weights. This is the integration by parts that carries the tan(latitude) terms without forming them.
- **Formed pointwise on the grid, then transformed back:** every product — $(f+\zeta)\nabla\psi$, $|\nabla\psi|^2$, $A\,\Phi_{\Pi\Pi}$, $S\,\zeta_n$, and the cross terms.

10.3 The vertical stencil

On the stretched Exner mesh the second derivative uses the standard three-point non-uniform formula. With $\delta_k^+ = \Pi_{k+1} - \Pi_k$, $\delta_k^- = \Pi_k - \Pi_{k-1}$ and $\delta_k^2 = \Pi_{k+1} - \Pi_{k-1}$,

$$\left.\frac{\partial^2 F}{\partial \Pi^2}\right|_k = b_k^{\mathrm{lo}} F_{k-1} + b_k^{\mathrm{c}} F_k + b_k^{\mathrm{hi}} F_{k+1}, \qquad b_k^{\mathrm{c}} = \frac{-2}{\delta_k^+ \delta_k^-}, \quad b_k^{\mathrm{hi}} = \frac{2}{\delta_k^+ \delta_k^2}, \quad b_k^{\mathrm{lo}} = \frac{2}{\delta_k^- \delta_k^2}. \tag{46}$$

It is exact for quadratics in Π, second-order on a uniform mesh and formally first-order on this one. Its rows sum to zero identically, so it annihilates a Π-constant — which is where the global-constant null direction of the system comes from.

10.4 Scaling of the unknowns and of the rows

The two equations differ by many orders of magnitude in units alone. Left as they are, a single Krylov tolerance on the combined vector is met almost entirely by the potential-vorticity row while the balance row stays far from converged, at a cost of order a third of a metre of geopotential height. The unknown column is therefore taken as $(\Phi,\ f_0\psi)$ with $f_0 = 10^{-4}\,\mathrm{s}^{-1}$, so that geostrophy makes the two comparable, and the equation rows as $(r_1,\ (f_0/\bar{S}_k)\,r_2)$ with $\bar{S}_k$ the level-mean static stability. This is a similarity transform: the solution is unchanged.

10.5 The gauge

The operator has an exact null space of dimension $n_{\mathrm{int}} + 1$:

- a constant added to ψ_n on *each* level. It carries no wind, and every term annihilates it.
- a *single global* constant added to Φ_n, annihilated because the folded vertical stencil's rows sum to zero and ∇^2 of a constant is zero. A per-level geopotential constant is *not* in the null space.

Both are removed inside the operator, at no extra cost.

The balance equation is a sum of divergences, so its $n = 0$ spherical-harmonic component vanishes identically for any input: that slot carries no information, and it corresponds precisely to the per-level streamfunction constant. It is therefore overwritten with the condition itself — the row returns ψ_n's mean and the right-hand side asks for zero — which pins each level's mean streamfunction.

The remaining constant is removed by a rank-one term on the potential-vorticity row, proportional to the column mean of Φ_n's $n = 0$ coefficient, with a coefficient chosen comparable to the operator's own diagonal.

On a limited-area domain a Dirichlet wall does both of these implicitly, and also absorbs any global imbalance between the area-mean potential vorticity and the boundary temperature. Globally that imbalance appears instead as a compatibility condition on the $n = 0$ column, whose removed magnitude is reported rather than discarded silently.

10.6 The preconditioner

Freeze the two coefficients at their per-level area means, A_k and S_k, and drop the cross and deformation terms. Spherical harmonics then diagonalise the horizontal part with $\lambda_n = -n(n+1)/a^2$, and each coefficient obeys a small system in the column alone:

$$\lambda_n \Phi - A\lambda_n \psi = r_1, \qquad A\,(T\Phi) + S\lambda_n \psi = r_2, \tag{47}$$

where T is the folded vertical second difference with unit coefficient. Eliminating ψ *exactly* leaves one tridiagonal problem per harmonic,

$$\left[\operatorname{diag}\left(\frac{A^2}{S}\right) T + \lambda_n I\right] \Phi = \frac{A}{S} r_2 + r_1, \qquad \psi = \frac{1}{A}\left(\Phi - \frac{r_1}{\lambda_n}\right). \tag{48}$$

This is the exact block factorisation of an approximate operator, not a block-diagonal approximation of the exact one: it costs the same and is strictly stronger. The reduced matrix is negative definite for every $n \geq 1$ because T is negative semi-definite and $\lambda_n < 0$, so there is no wavenumber at which the preconditioner degrades. The small inverses are dense and are precomputed once per event for each n. The $n = 0$ block is handled separately, using the rank-one corrected operator for the geopotential and reading the streamfunction mean straight off the balance row.

The construction fails loudly if any $A_k \leq 0$ or $S_k \leq 0$, which is the same ellipticity requirement as the system itself.

10.7 The linear solve

Each piece is one preconditioned Krylov solve. The default is GMRES with restart 60 and a relative tolerance of 10^{-8}. The iteration budget is stated as a number of operator applications (400 by default) and converted internally to restart cycles. A solve is reported as converged only if the solver says so *and* an independently recomputed $\|b - Mx\|/\|b\|$ is below $\max(10\,\mathrm{rtol}, 10^{-10})$, which at the defaults is 10^{-7}. A solve that exhausts its budget returns its best iterate and is marked unconverged rather than raising, so one difficult piece does not end a run.

10.8 The nonlinear solve

The total balanced state is needed before any piece can be defined, because the perturbation ψ' that the operator is frozen at is measured from it. It is found by Newton's method, and it needs no solver of its own: because both nonlinearities are quadratic, the Jacobian of the residual *is* the linearised operator (34)–(35) evaluated at the current iterate. The same operator and the same preconditioner are reused.

One step is worth spelling out. The residual is evaluated *through* the same code path as the Jacobian, using

$$F(x) = J_{x/2}[x] + F(0), \tag{49}$$

which holds exactly for a quadratic system. Evaluated instead from the pristine equations, the residual and the Jacobian describe slightly different systems wherever a coefficient has been floored or tapered, and the iteration stalls after roughly halving the residual.

The remaining ingredients:

- **Forcing term.** The linear tolerance for step k is $\min(0.1,\ \max(\mathrm{rtol},\ 0.1\,\|r_k\|/\|r_0\|))$, normalised by the first residual. Step 0 therefore uses 0.1.
- **Line search.** Backtracking by halves, up to eight times, accepting the first step with $\|F(x + s\delta)\| \leq (1 - 10^{-4}s)\|F(x)\|$. If all eight fail, $s = 2^{-8}$ is taken and the iteration continues.
- **Stopping rule.** The largest geopotential increment over the interior grid below $0.981\ \mathrm{m}^2\ \mathrm{s}^{-2}$, which is g times 0.1 m, with a cap of 20 steps. A run that hits the cap is recorded, not raised, and reports how far short it stopped in metres — a Boolean cannot distinguish an iteration asymptoting a hair above tolerance from one stalled an order of magnitude above it, and the difference decides whether the result is usable.

- **Boundary levels** are rebuilt hydrostatically after every step: Φ from θ directly, ψ from (43).

Uniqueness of the converged state is an empirical claim, not a proved one. Davis and Emanuel (1991, p. 1935) report that as long as the potential vorticity was everywhere positive their iteration consistently converged, and that tests with different initial guesses showed the solution obtained was apparently unique, or at least that other solutions were unreachable. That is the standing of the statement here as well.

10.9 Cost

Measured at truncation 63 on a 96×192 solver grid with nine levels: one operator application takes about 18 ms; a preconditioned solve of a manufactured problem converges in 33 GMRES iterations to 6×10^{-11}; a real event takes about two minutes end to end, its Newton iteration converging in about a dozen steps and each of its nine pieces in 35–51 Krylov iterations.

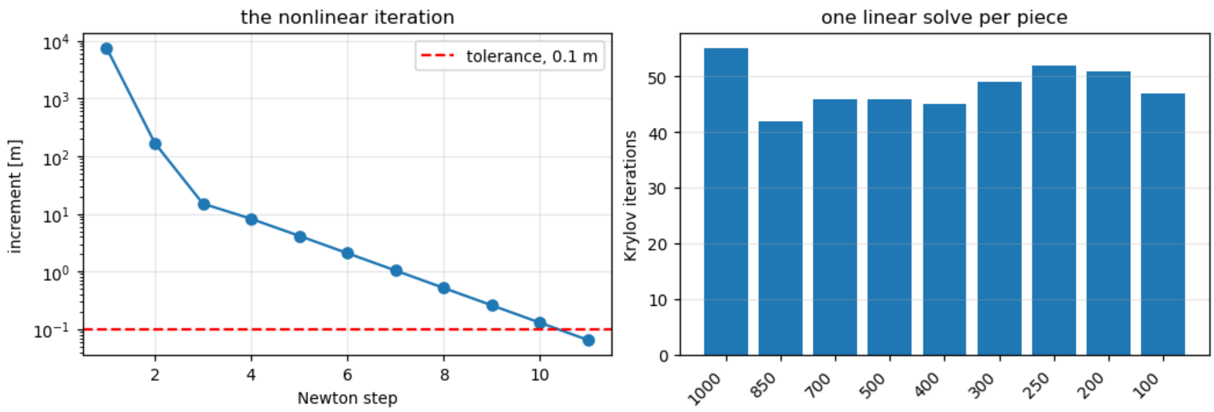

Figure 5: The nonlinear iteration and the Krylov counts. Left: the Newton increment, in metres of geopotential height, against the 0.1 m tolerance. Right: the number of Krylov iterations taken by each of the nine pieces, all solved against one frozen operator.

11 The wind induced by each level

This is the result. Each panel of Figure 6 is the rotational wind at 250 hPa that would exist if the only anomaly in the atmosphere were the one on that level, with everything else at its climatological value. Seven of the nine sources are interior potential-vorticity anomalies; two are boundary potential-temperature anomalies.

Two things to know before reading amplitudes off such a figure.

The two outermost levels of every piece are hydrostatic extensions, not solved levels. Their values are the level next to them plus a boundary temperature term. Interior levels are the solved ones. In the current implementation the streamfunction's extension at those two rows uses the geopotential's ghost rather than (43); the interior levels and the closure are unaffected, because the discrepancy is linear in θ and cancels in the sum, but an amplitude quoted on the 1000 or 100 hPa row of a single piece should be read with that in mind.

A large piece is not necessarily a relevant one. The top-boundary piece in particular can reach tens of metres per second while projecting almost nothing onto the event, because a planetary-scale temperature anomaly at 100 hPa has a Rossby depth exceeding the domain and its response is deep and nearly barotropic. It largely cancels against the upper-level pieces. Amplitude and projection are different questions and should be reported separately.

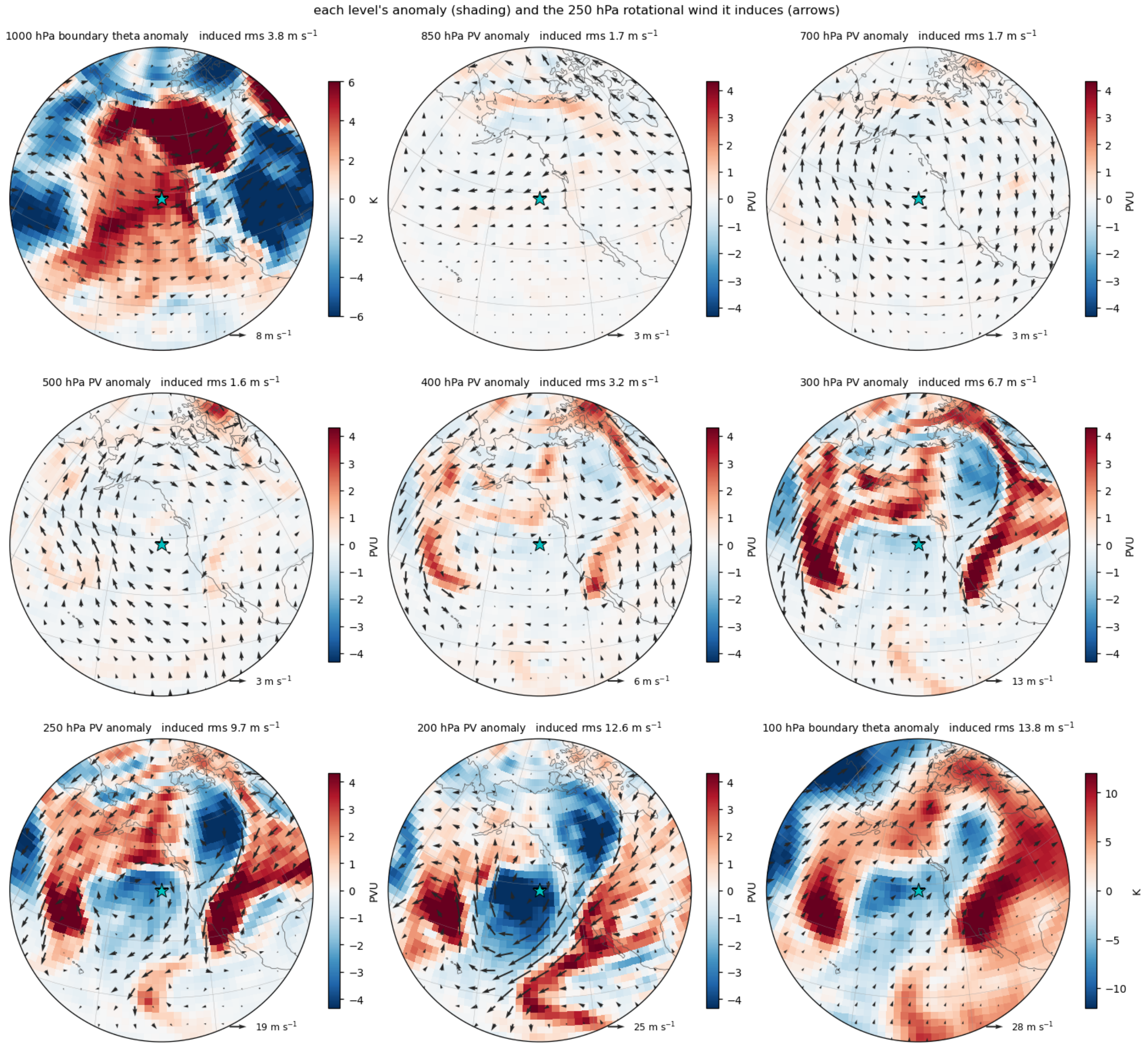

Figure 6: The wind at 250 hPa induced by each level. Shading is the source that drives the piece — potential vorticity on the seven interior levels, potential temperature on the two boundaries — and the arrows are the rotational wind it induces at 250 hPa. The arrow scale is set panel by panel and written on each one.

12 Grouping the pieces

Nine panels is more than the eye can hold, and the usual question is coarser: how much of the upper-level flow came from below, and how much from the upper troposphere? Because the pieces are linear in their sources, groups are simply sums of pieces, exactly:

group	sources
surface	1000 hPa boundary potential temperature
lower	potential vorticity at 850, 700, 500 hPa
upper	potential vorticity at 400, 300, 250, 200 hPa and the 100 hPa boundary

The three groups exhaust the sources, so they sum to the same thing the nine panels do. Any other partition of the nine works the same way, and it is done after the solve, at no cost.

A group may also be defined by something other than a level list — by horizontal scale, say, or by a mask around the event. When it is, the split must be applied to the *source* and it must be additive: the parts must sum to the whole source. A multiplicative mask applied to a solution does not close. For the same reason, a group defined by scale must split every source it carries, including the boundary temperature, not only the interior potential vorticity.

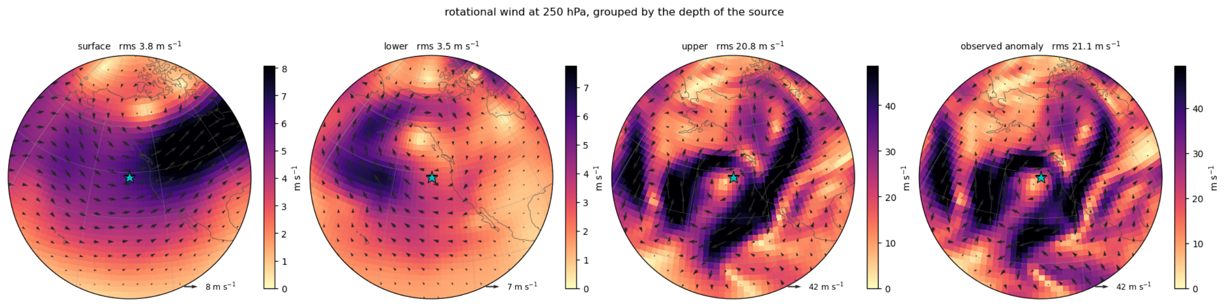

Figure 7: The pieces grouped into surface, lower and upper contributions, beside the observed anomalous rotational wind at 250 hPa. Colour and arrow scales are set panel by panel; the numbers in the titles are what to compare across panels.

13 Closure, and what is left over

Two checks close the calculation, and they are different in kind.

The arithmetic check. The nine pieces must reproduce the inversion of all the sources at once. This is not an approximation to be assessed: it follows from the four facts of Section 7 and holds to the solver's tolerance, measured at the level of 10^{-8} relative on a real event. If it did not hold, the decomposition would mean nothing, so it is worth checking on every run rather than trusting.

The physical check. The sum of the pieces against the *observed* anomalous rotational wind. What is left over is the part the balance equations do not represent: the divergent and unbalanced flow. It is not an error and it is not a wall response — there is no wall. Its size is a property of the event and is worth reporting.

The distinction to hold on to: the pieces sum to the all-sources linear solve; they do not sum to the difference between the balanced total state and the observed mean state. The gap between those two is exactly the physical residual above, and it is stored under its own name rather than distributed over the pieces.

14 The same calculation at 79 N

Nothing in Sections 2 to 13 refers to where the event is. The operators are global, the grid has no pole row, no metric factor is divided by, and there is no box to draw. An event at 79 N, half of whose surroundings lie across the pole, is inverted by the same calculation in the same number of Newton steps as one at 43 N.

Two practical points about output near the pole.

The eastward and northward *components* of a wind are undefined exactly at the pole — their values depend on the meridian one approaches along — even though the wind itself is perfectly

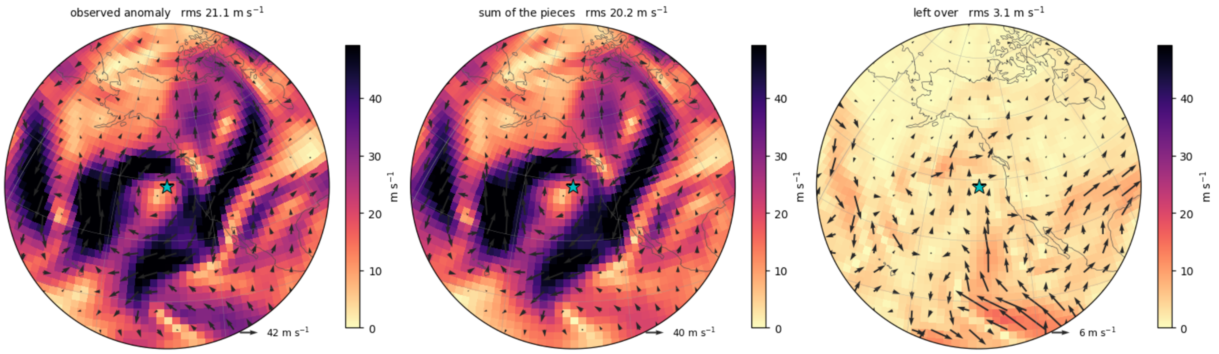

Figure 8: The sum of the pieces, and what is left over. Left: the observed anomalous rotational wind at 250 hPa. Centre: the sum of the nine pieces. Right: the difference, which is the unbalanced and divergent part.

well defined there. Output rows where $|\cos\varphi| \leq 10^{-8}$ are therefore returned as missing rather than filled with a number that pretends otherwise.

For cropping a patch around a near-polar event, a rotated frame is available: the frame is rotated so the event sits at the pole of the projection and a disc of fixed great-circle radius is cut. A geographic latitude–longitude box centred at 79 N is largely empty because those rows lie past the pole; the rotated patch is complete, with the same peak amplitude.

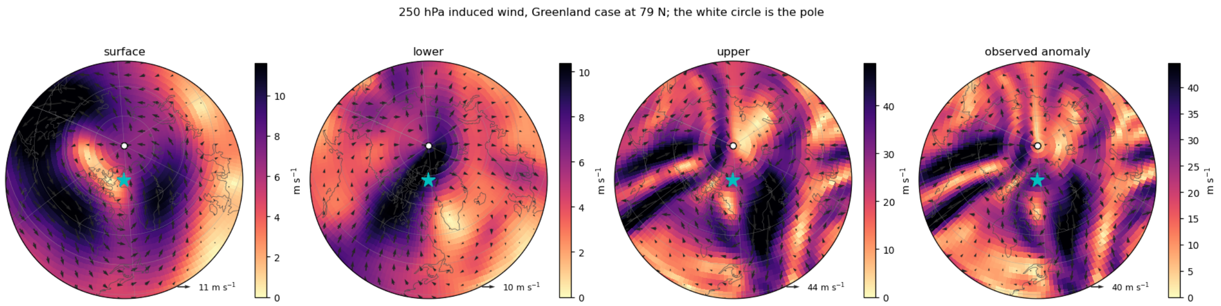

Figure 9: The same calculation for an event at 79 N. The white circle is the North Pole, which is an ordinary point inside the picture.

15 The choices, and what they cost

This section collects every place where the solved system is not the system as posed, with numbers.

15.1 Modifications in a band around the equator

Two of them, both described in Section 9 and both on by default.

what	default	effect
Coriolis floor φ_0	12°	$f^*/\lvert f\rvert = 1.52$ at 10.5 N, 1.17 at 20 N, 1.08 at 30 N, 1.04 at 45 N
taper band	5°–20°	74% of the reference vorticity removed at 10.5 N, nothing removed above 20 N

Inside that band the frozen operator is the exact finite-increment operator of a *modified* system, not of the equations as posed. Measured on a global synthetic case that needs no taper at all, turning the taper on leaves about half a metre of equivalent geopotential height in the converged state; with it off on a mirrored state, the kink at the equator acts as a vortex sheet and rings.

A solution quoted anywhere in that band must be checked against a run with a smaller floor and a narrower taper.

That check is a requirement, not a suggestion, and its outcome is not the one that might be expected. Narrowing the band makes the answer worse, not better. Measured over the southern third of an event patch — the part that reaches into the band — the residual against the observed anomalous rotational wind is:

configuration	residual [m s^{-1}]
taper 5°–20°, floor 12° (default)	3.30
taper 2°–8°, floor 4°	4.19
no taper, floor 4°	4.67

The default is the best of the three. A narrow band leaves the kink at the equator inside the operator's coefficients, and a small floor leaves ∇f^* nearly discontinuous there; both radiate into the subtropics. The rule is therefore to run both configurations and report both, not to treat the smaller floor as the truthful one.

One further inconsistency inside the band, stated for completeness. With the taper on, the balance row's reference streamfunction is rebuilt as ∇^{-2} of the tapered vorticity, while the cross-term coefficients of the potential-vorticity row are taken from the untapered spectra and then ramped to zero. Because ∇^{-2} is non-local, those two reference flows are not literally the same field, inside the band or outside it. It does not break the sum-closure — every piece shares the same operator — and for the same reason the nonlinear residual cannot see it.

15.2 Floors that keep the operator elliptic

name	default	applies to	note
`avo_min`	10^{-6} s^{-1}	$A = f^* + \nabla^2\tilde{\psi}$	absolute vorticity
`stb_min`	10^{-4}	$S = \tilde{\Phi}_{\Pi\Pi}$	static stability

The floors are stated in SI deliberately: in a non-dimensional scaling the same literal changes physical meaning with resolution and level count.

Two application rules are provided. The default, `parity`, compares against one tenth of the floor and then assigns the floor, so a value between the two is left alone; `clean` compares against the floor itself. The clamped area fraction per level is published with every run, because the clamps activate under strong anticyclones and blocks — which is exactly where the science is, so a run should always be able to show how much of its domain was propped up.

(The configuration also declares a `deformation_margin`; it is not read anywhere and does nothing.)

15.3 Potential-vorticity floors

name	default	where
`qmin_total`	0.01 PVU	the total (nonlinear) inversion
`qmin_pieces`	10^{-5} PVU	event and mean states *separately*, before differencing

The floor gives the added amount back to the unclamped points, weighted by area, so the volume integral is preserved; whatever cannot be given back is reported as a conservation slack rather than absorbed. Under the `parity` rule the redistribution is a single pass; under `clean` it iterates.

The floor is not linear, so the source of a piece is the anomaly of two separately floored fields, not the floored anomaly. That is the reason for the ordering, and it is why `qmin_pieces` is far smaller than `qmin_total`: the piecewise system is linear once frozen and does not need positivity, whereas the nonlinear iteration does.

15.4 What "exact" means, and where it stops

The claim of Section 7 is exactness *of the operator that is actually built*. Two things modify that operator — the floors above and the equatorial taper — and wherever either bites, the operator is the exact finite-increment operator of a modified system. Everywhere else it is the exact finite-increment operator of (28)–(29).

The closure of the pieces is unaffected by either, because all pieces share the same operator. That is precisely why the taper is applied to coefficients only.

15.5 Other things worth knowing

- **Below ground.** Where a pressure level lies under the surface, temperature and wind are carried down and the height is continued hydrostatically. Those points are filled, not observed.
- **The mirrored scaffold.** South of the data's own southernmost row, every value is a smooth, correctly scaled reflection of the north and entirely fabricated. Output rows there are returned blank, so that a patch reaching into them cannot enter a composite as if they were observations.
- **Input guards.** A run stops rather than continues if the geopotential decreases with height at more than 1% of points (which catches a top-down level axis), or if potential temperature decreases with height at more than 5% of points (which catches θ supplied in place of T).
- **Column averages** divide by the weight of the levels that were actually valid at each point, never by the full count. Filling a gap with zero and dividing by everything pulls the answer towards zero while keeping it finite, which is far harder to notice than a missing value.
- **The residual is not an error.** It is the unbalanced and divergent flow, and it belongs to the event.

16 Running it

16.1 Installation

```
pip install -e .
```

The requirements are NumPy, SciPy, threadpoolctl, xarray and netCDF4; matplotlib and cartopy are needed only for the figures. SciPy 1.12 or newer is required, because the Krylov drivers are called with the modern `rtol` keyword.

16.2 Input contract

Each state is four arrays of shape $(n_{\mathrm{lev}}, n_{\mathrm{lat}}, n_{\mathrm{lon}})$ covering the northern hemisphere:

`lat`	ascending, 0 to 90 N
`lon`	ascending, starting at 0
`level`	descending in pressure, 1000 hPa first
`z`	geopotential height [m] — not geopotential
`t`	temperature [K] — not potential temperature
`u`, `v`	wind components [m s^{-1}]

A climatology on the same grid and the same levels is required alongside the event.

16.3 One event, from Python

```
from pvinv_sph.config import InversionConfig, MirrorConfig
from pvinv_sph.levels import build_levels
from pvinv_sph.passd import invert_pieces
from pvinv_sph.prepare import prepare_state
from pvinv_sph.sht import SHT, gaussian_grid
from pvinv_sph.sphere import SphereOps
from pvinv_sph.winds import rotational_wind_stack

levels = build_levels("NL9")
solver = SphereOps(SHT(gaussian_grid(96, 192), lmax=63))
cfg    = InversionConfig(mirror=MirrorConfig(blend=True))

event = prepare_state(z_ev, t_ev, u_ev, v_ev, lat, lon, levels, solver)
clim  = prepare_state(z_cl, t_cl, u_cl, v_cl, lat, lon, levels, solver)

result = invert_pieces(solver, levels, clim, event, cfg=cfg)

# the rotational wind induced by the 300 hPa potential-vorticity anomaly
u, v = rotational_wind_stack(solver, result.pieces["300"].psi_spec)
```

`result.pieces` is keyed by pressure in hPa: `"1000"` is the surface potential-temperature piece, `"850"` through `"200"` the interior potential-vorticity levels, and `"100"` the top boundary. `result.diagnostics` carries the Newton history, the Krylov count per piece, and the clamp and floor reports. The sum of the pieces is `result.summed_psi()`, to be compared against `all_sources_inversion` for the closure check of Section 13.

For a complete event including the patch cropping and the output arrays, use `invert_event` from `pvinv_sph.pipeline`. It takes the two states, the event centre and the patch half-widths, and returns the arrays ready to save.

16.4 A catalogue, from the command line

```
pvinv-sph catalogue.csv --out-dir results/ --workers 32
```

The catalogue is a CSV whose columns are `event_id`, `state_path`, `state_index`, `clim_path`, `clim_index`, `lat` and `lon`. They are checked before the run starts. Events are run in parallel with one BLAS thread each — the arrays are small enough that threading them buys nothing and the spin waits cost everything — and a failure is recorded per event rather than raised, so one pathological mean state does not end a run of thirty thousand.

The flags that matter for the approximations of Section 15:

flag	default	meaning
`-f-floor-deg`	12.0	latitude scale of the Coriolis floor
`-blend-south`	5.0	equatorward edge of the coefficient taper
`-blend-north`	20.0	poleward edge of the coefficient taper
`-no-blend`	—	skip the taper entirely
`-newton-max-steps`	20	cap on the nonlinear iteration
`-solver-nlat`, `-solver-nlon`	128, 256	the Gaussian solver grid
`-lat-half`, `-lon-half`	30.0, 60.0	output patch half-widths
`-rotated-track`	—	also write the rotated-frame patch

A run that hits the Newton cap is not automatically a failure: check the reported final increment in metres before treating it as one.

16.5 The walkthrough, and the tests

Every figure in this document is produced by the notebook in `examples/01_walkthrough/`, which runs one inversion from beginning to end on a sample event that ships with the package. Regenerate them with

```
python examples/01_walkthrough/build.py
jupyter nbconvert --to notebook --execute --inplace \
    --ExecutePreprocessor.timeout=3600 \
    examples/01_walkthrough/01_walkthrough.ipynb
```

and run the test suite with

```
PYTHONPATH=src python -m pytest tests -q
```

The tests are the verification ladder for everything asserted above: that the vertical stencil is exact for quadratics in Π and that its rows sum to zero; that the boundary folding and the boundary source are consistent; that the balance term reproduces the closed form for solid-body rotation and is invariant under rotating a vortex to the pole; that the spectral round trip is exact; and that the pieces sum to the all-sources solve.

References

[1] Charney, J., 1955: The Use of the Primitive Equations of Motion in Numerical Prediction. *Tellus*, **VII**(1), 22–26.

[2] Davis, C. A., and K. A. Emanuel, 1991: Potential Vorticity Diagnostics of Cyclogenesis. *Monthly Weather Review*, **119**(8), 1929–1953.

[3] Davis, C. A., 1992: Piecewise Potential Vorticity Inversion. *Journal of the Atmospheric Sciences*, **49**(16), 1397–1411.